



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	24 NOVEMBER 2022
GRADE	11	MARKS	170
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	3 HOURS
CLASS			

INSTRUCTIONS:

1. This paper consists of 26 pages, a yellow Answer Booklet of 11 pages (2-10) and an Appendix of coloured images (25-26). Please check that your question paper is complete.
2. This question paper consists of four questions.
3. Question 1 must be answered **in the Yellow Answer Booklet provided**.
4. Questions 2, 3 and 4 must be answered **on the folio paper provided**.
5. Start each question on a new page.
6. Read the questions carefully.
7. Number the answers exactly as the questions are numbered.
8. Use the total marks that can be awarded for each of Questions 1, 2, 3 and 4 as an indication of the detail required.
9. It is in your own interest to write legibly and to present your work neatly.

QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	70	
2	30	
3	30	
4	40	
TOTAL: 170		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing only the letter of the most correct answer in the box below. **[10]**

1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8	1.1.9	1.1.10

- 1.1.1. Which one of the following methods is regarded as a direct technique to determine population size?

- A. Simple sampling
- B. Census
- C. Mark recapture
- D. Quadrat method

- 1.1.2. A decrease in the size of a population is due to...

- A. Migration
- B. Increased immigration
- C. Increased emigration
- D. Natality

- 1.1.3. The resource partitioning observed in an ecosystem effectively reduces...

- A. The ability of closely related species to interbreed.
- B. Competition between closely related species by reducing overlap of ecological niches.
- C. Predatory instinct of related species.
- D. The wastage of food resources.

- 1.1.4. The use of antibiotics is an effective treatment for...

- A. Bacterial and viral infections
- B. Bacterial infections only
- C. Viral infections only
- D. Neither bacterial nor viral infections

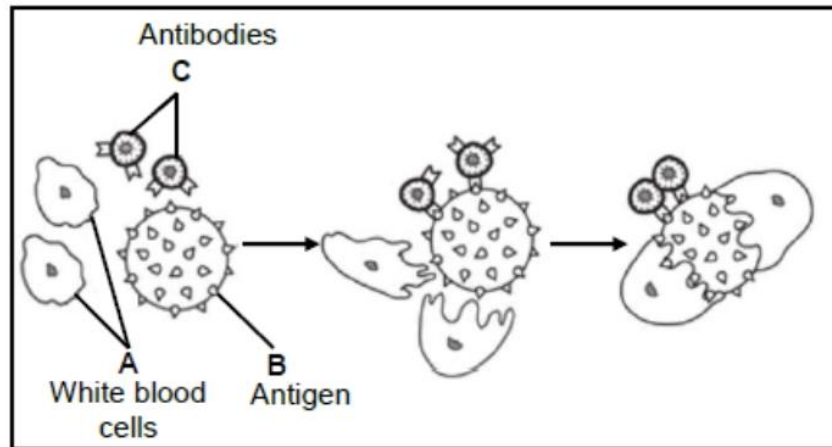
- 1.1.5. Which part of the neuron transmits impulses towards the cell body?

- A. Dendrite
- B. Myelin sheath
- C. Axon
- D. Receptor

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1.1.6. The diagram below illustrates activities taking place in a human body.

Based on the diagram, vaccinations usually stimulate the body to produce more of...

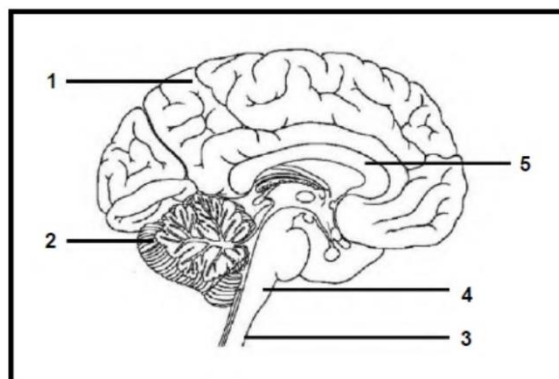


- A. Structure A, only.
- B. Structure B, only.
- C. Structures A and C, only.
- D. Structures A, B and C.

1.1.7. Which statement correctly describes fungi and protozoans?

- A. Fungi and protozoans are all unicellular.
- B. Fungi and protozoans all have chloroplasts.
- C. Fungi and protozoans are eukaryotic.
- D. Fungi and protozoans all have cell walls.

QUESTIONS 1.1.8.- 1.1.10 ARE BASED ON THE DIAGRAM BELOW.



1.1.8. Which part controls balance and equilibrium?

- A. 1
- B. 2
- C. 4
- D. 5

1.1.9. Which one of the following is a possible symptom in a person whose part 4 is not functioning well?

- A. Irregular heartbeat
- B. Memory loss
- C. Slow response to the stimulus
- D. Loss of sight

1.1.10. Part 1 is more commonly known as the...

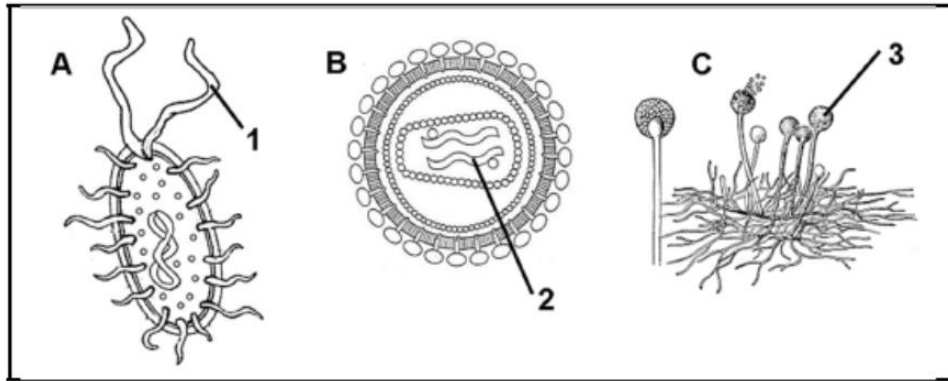
- A. The occipital lobe
- B. The temporal lobe
- C. The parietal lobe
- D. The frontal lobe

1.2. Select the name/ or term in Column II which best matches the description in Column I. Write the letter of the matching name/term in the appropriate space provided between the brackets on the very left-hand side. Each letter may only be used once. **[10]**

<u>Column I</u>		<u>Column II</u>
[]	An indication of the variety of living things found in an environment.	A succession
[]	Process of gradual change of an ecological community over time.	B carrying capacity
[]	Linked food chains showing the energy flow in an ecosystem	C food web
[]	A group of organisms occupying the same area that can breed freely with one another.	D biotic factors
[]	Organisms that hunt and kill their food.	E abiotic factors
[]	Dividing vegetation into vertical layers.	F biodiversity
[]	Non-living components in the environment.	G community
[]	Area in which living and non-living things interact with each other.	H stratification
[]	Living components in the environment.	I ecosystem
[]	The number of individuals an area can sustain.	J population
		K predator

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1.3. Study the diagrams below of different microorganisms and answer the questions that follow.



1.3.1. Name the group of organisms to which A and C belong:

A- _____ C- _____ (2)

1.3.2. Provide labels for the following structures:

1- _____
 2- _____
 3- _____ (3)

1.3.3. Give the letter of the organism that:

a. Is not considered to be living: _____ (1)
 b. Is eukaryotic: _____ (1)

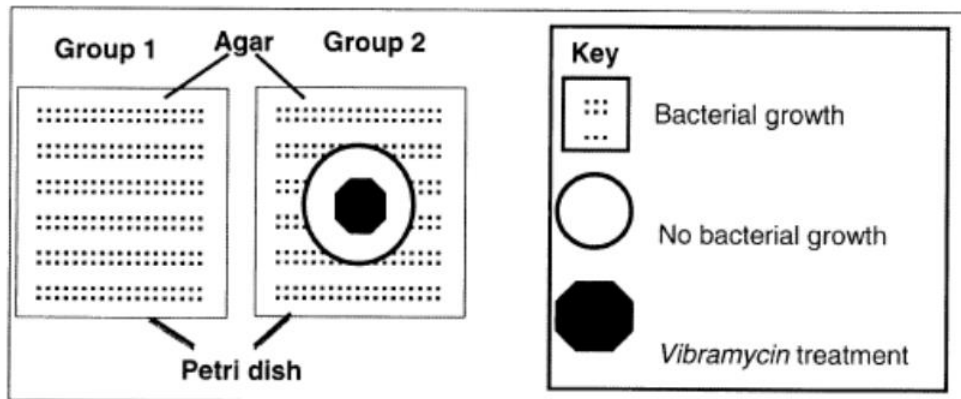
1.3.4. What is the shape of the organism in A? (1)

1.3.5. In the space below draw a labelled diagram of structure 3. (5)

1.4. Grade 11 learners wanted to investigate whether a new antibiotic, called Vibramycin, could restrict the production of a population of bacteria which was resistant to other antibiotics.

Their investigation was set up as follows:

- The bacteria were cultured in the same type of agar medium in 20 petri dishes of the same size and then divided into 2 groups:
 - **Group 1:** 10 petri dishes were given no treatment
 - **Group 2:** 10 petri dishes were treated with Vibramycin of the same concentration.
- The petri dishes were then incubated under the same conditions and examined for bacterial growth.
- The diameter of the area where no bacteria grew was measured for each petri dish.



1.4.1. What is the dependent variable in the above investigation? (1)

1.4.2. Identify THREE factors that should be kept constant during the investigation, other than those mentioned above. (3)

1.4.3. Why is more than one petri dish used in each group? (1)

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1.4.4. What was the aim of Group 1 in the investigation?

(1)

1.4.5. A similar experiment was done in which the cultured bacteria were divided into 5 groups. These groups were then treated with different concentrations of Vibramycin.

a. Formulate a hypothesis for the above investigation.

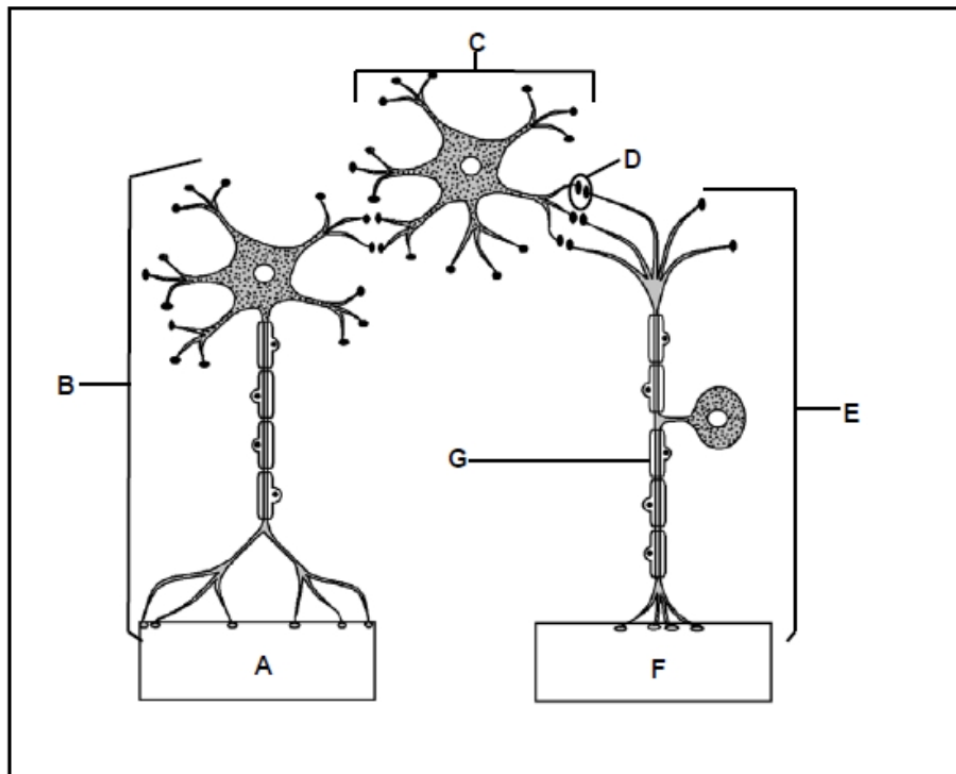
(2)

b. State the independent variable for this investigation.

(1)

[9]

1.5. The diagram below represents a possible 'path' followed by an impulse when a person touches a hot plate.



1.5.1. Name the 'path' illustrated above.

(1)

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1.5.2. What is the importance of the path mentioned in QUESTION 1.5.1.? (2)

1.5.3. Identify the type of neuron represented by: (3)

- a. B: _____
b. C: _____
c. E: _____

1.5.4. Give the LETTER only of the part that represents the: (2)

- a. Receptor: _____
b. Effector: _____

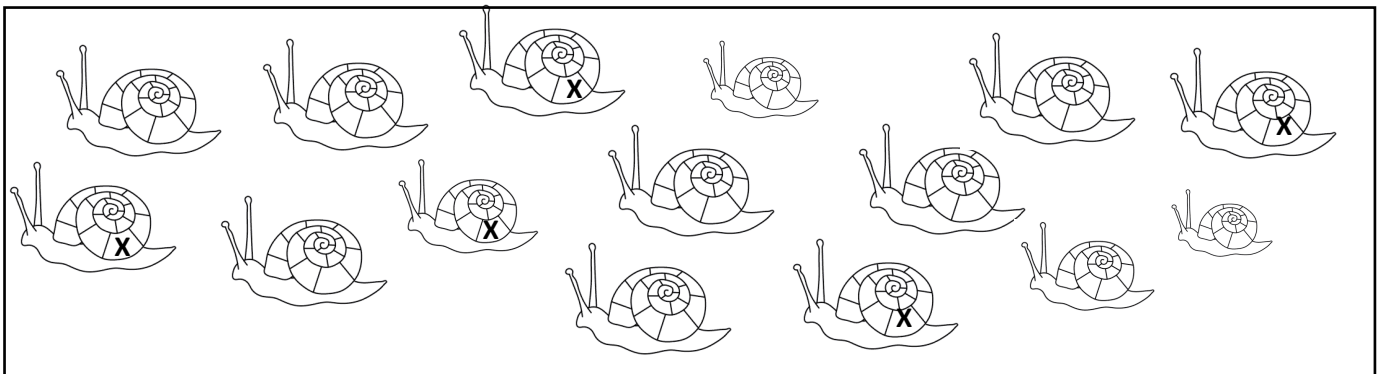
1.5.5. Give the LETTER and NAME of the: (4)

- a. Region where the impulse is transmitted chemically: _____
b. Part that has an insulating function: _____

1.5.6. What would happen if Neuron B were to be cut? (2)

[14]

- 1.6. In an investigation to determine the size of the snail population in a garden, snails were collected at random. Each one was marked with an X and then returned to the garden. After one week, a second sample of snails was collected. The diagram shows the snails that were caught during the second selection.



1.6.1. What is the name of the technique that was used to determine the population of the snails in this investigation? (1)

1.6.2. Would the method in QUESTION 1.6.1. be classified as a direct or indirect method. (1)

1.6.3. Give a reason for your answer to QUESTION 1.6.2. (2)

1.6.4. Would you say the technique used to determine the snail population in the garden would be the most accurate? Give a reason for your answer. (2)

1.6.5. Using a formula, it is determined that there are 60 snails in the population. How many snails were there in the first sample? (3)

1.6.6. Name TWO precautions that needed to be taken when performing the technique used in the above population estimation. (2)

1.7. Read the information given below on plants that grow in the grassland biome.

Plant A	• Live for a few years • Fast growing • Grows in low and high nutrient soils • Roots not widespread
Plant B	• Live for many years • Slow growing • Grows in high nutrient soils • Has deep and widespread roots
Plant C	• Germinates quickly • Fast growing • Grows in low nutrient soils • Also grows in disturbed areas

Decide where each plant fits in the process of ecological succession. Write the letter of the plant in the correct stage of succession in the textboxes below.



[3]

TOTAL QUESTION 1: 70 MARKS



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LIFE SCIENCES PAPER 1: QUESTIONS 2, 3 AND 4

Name: _____

Key: _____

Time: 3 hours

Marks: 170

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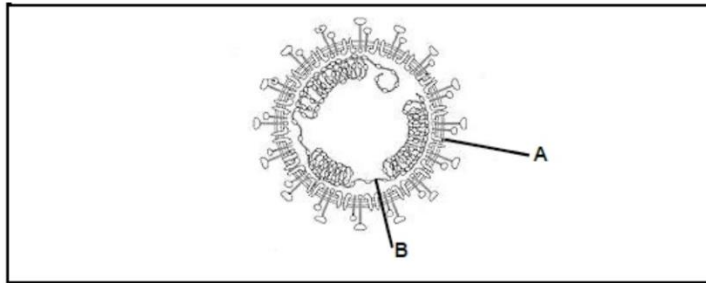
Examiner: Mrs S Stegmann

Moderator: Mrs B Zajac

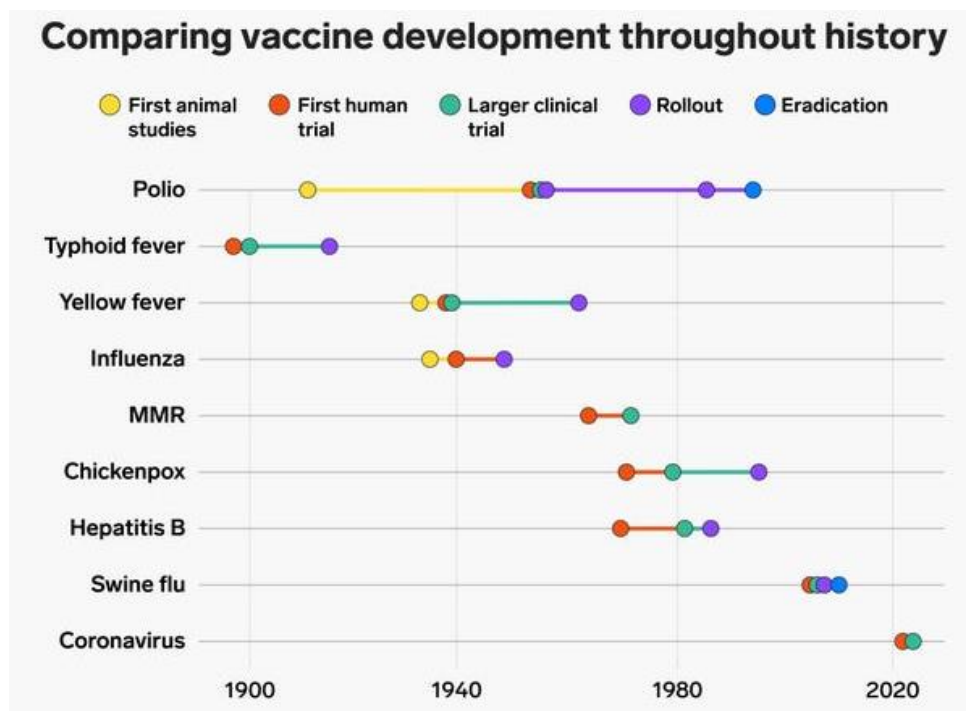
Question 2:

- 2.1. The novel corona virus, named COVID-19, escalated to a pandemic level during 2020. Scientists around the world raced to produce a vaccine to help fight the virus. However, it would take months before a vaccine could be released, once it had been discovered.

Below is the virus that caused our world to change.



- 2.1.1. Provide labels for parts A and B respectively. (2)
- 2.1.2. Why were antibiotics not prescribed to fight against the virus? (2)
- 2.1.3. Explain how a vaccine works to protect us from a disease. (3)
- 2.1.4. Why would it take so long for a vaccine to be released? (2)
- 2.1.5. The image below (*Appendix*) shows different vaccines' development throughout history.

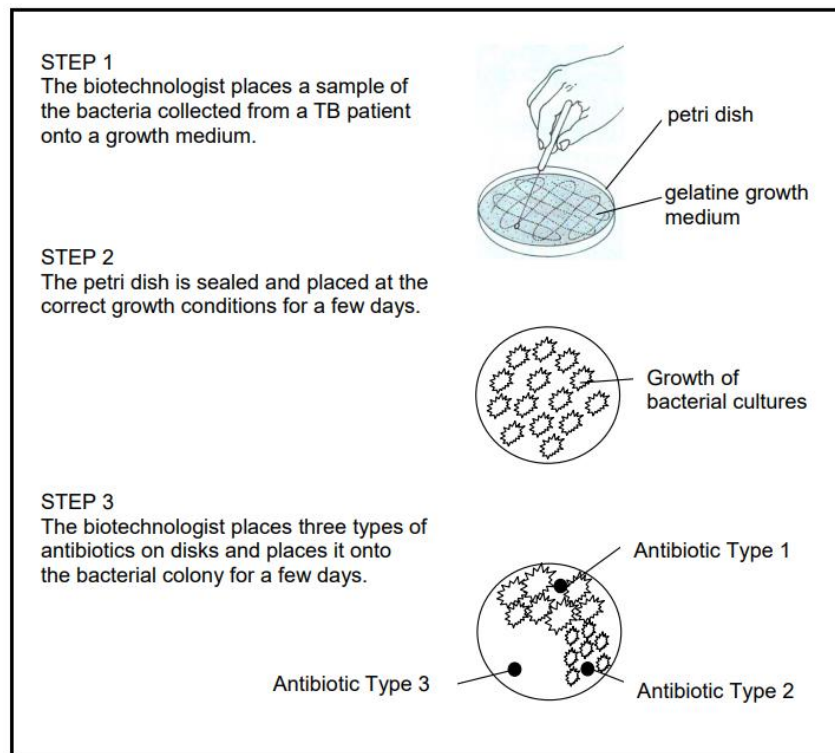


- 2.1.5.1. How long did it take the yellow fever vaccine to be rolled out from the first studies? (2)

- 2.1.5.2. Which disease have been eradicated (completely removed) since their vaccination was released? (2)
- 2.1.5.3. Describe the trend from the polio vaccination process to that of the swine flu. (2)
- 2.1.5.4. Provide a reason for your answer to QUESTION 2.1.5.3. (2)
- 2.1.5.5. Why are the cases of viruses such as the corona virus lowered in warmer months? (2)

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- 2.2. A biotechnologist investigates the effectiveness of antibiotics on the treatment of tuberculosis (TB) and follows the steps shown below (*Appendix*) :



- 2.2.1. Provide an aim for the above investigation. (2)
- 2.2.2. What type of microorganism causes TB? (1)
- 2.2.3. From the results in STEP 3 provide a conclusion for the investigation. (3)
- 2.2.4. How is tuberculosis treated? (2)
- 2.2.5. The incidence of TB is much higher in low income groups than in higher income groups. Suggest why this is so. (3)

[11]

Total Question 2: 30 marks

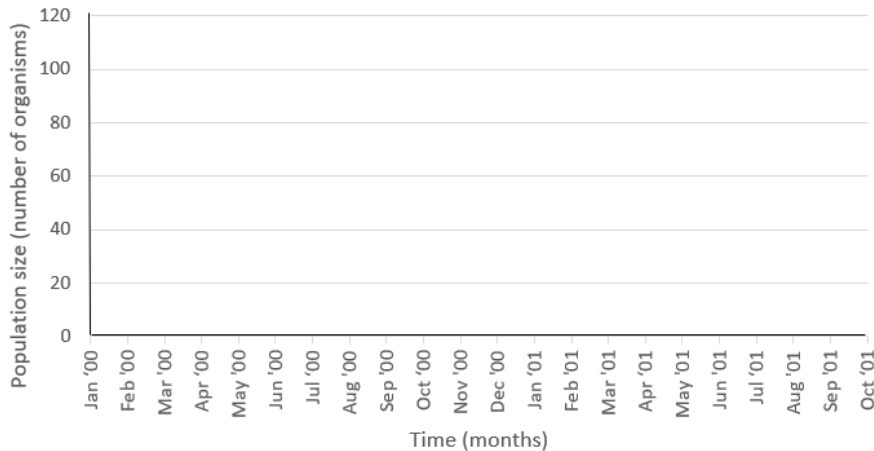
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Question 3:

- 3.1. The number of lady birds and green flies in an ecosystem were counted periodically from January 2000 ('00) to October 2001 ('01) and recorded below:

Time (Month)	Number of lady birds	Number of green flies
January '00	30	20
April '00	10	70
July '00	36	110
October '00	78	50
January '01	20	30
April '01	36	48
July '01	60	80
October '01	78	30

- 3.1.1. **Redraw** the set of axes below **to scale** and display the above information in a **double line graph**. (5)

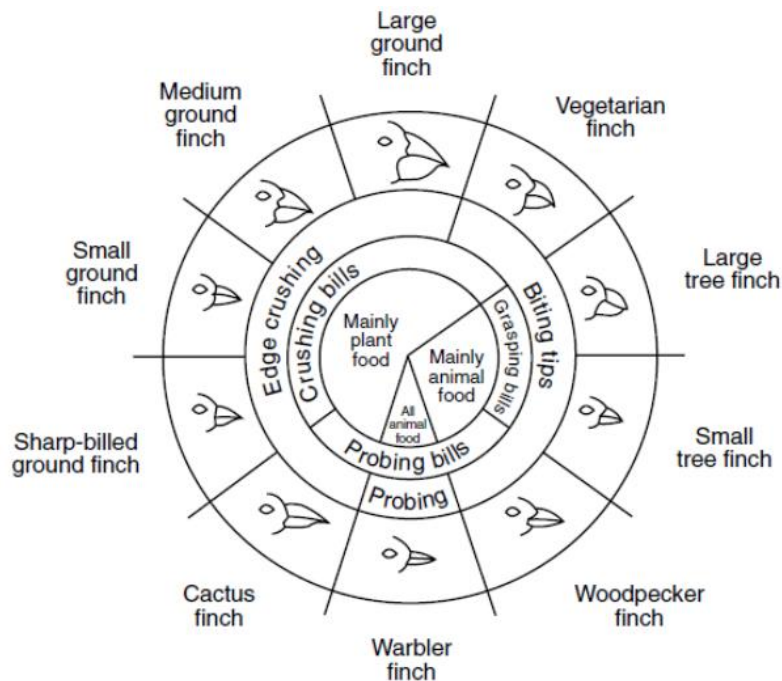


- 3.1.2. How often were the populations of both organisms counted? (1)
- 3.1.3. What type of relationship is displayed between the ladybirds and green flies? (1)
- 3.1.4. From the graph 3.1.1. determine the role played by the ladybirds in this relationship. (1)
- 3.1.5. Provide TWO reasons for your answer to QUESTION 3.1.4. (2)
- 3.1.6. A relationship similar to this one is shown between great white sharks and seals. Explain how both of these populations are benefitted from this relationship. (4)
- 3.1.7. Define the term 'carrying capacity'. (2)

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3.2. Fourteen finch species, with beaks of different shapes and sizes, originated from one common ancestor by means of evolution. The different shapes and sizes of their beaks determine the type of food they eat. Study the diagram below and answer the questions that follow.



3.2.1. Name the type of competition displayed above. (1)

3.2.2. What structural feature determines the different species of finch? (1)

3.2.3. In each of the following, write only the LETTER that corresponds: (2)

a. Which two finches will compete the least with each other for food:

- A small ground finch and large ground finch
- B large ground finch and sharp-billed ground finch
- C small tree finch and medium ground finch
- D vegetarian finch and small ground finch

b. The only exclusively carnivorous finch has a bill that is adapted as follows:

- A only probing
- B probing and edge crushing
- C probing and biting tips
- D biting tips and edge crushing

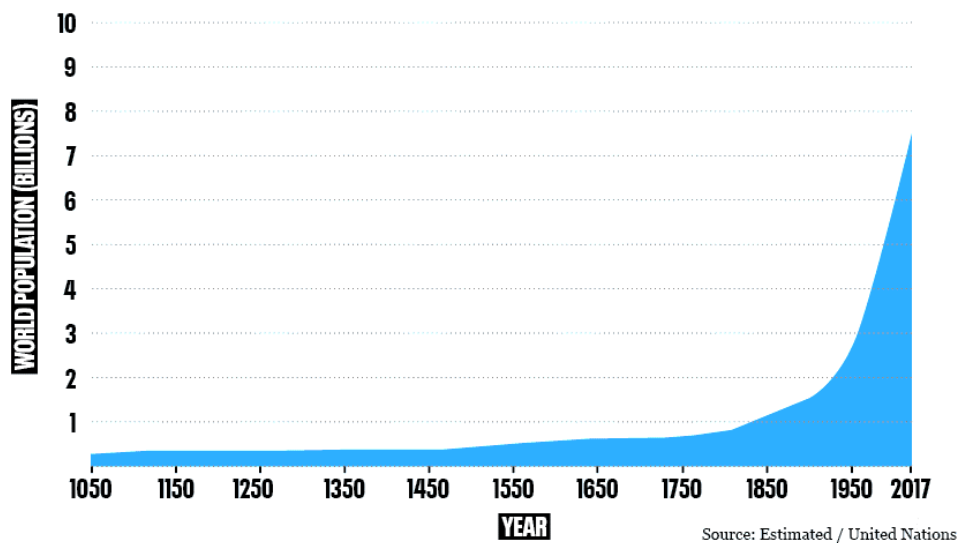
3.2.4. Would the large tree finch and large ground finch be able to co-exist on the same island? Give a reason for your answer. (2)

3.2.5. Which finch species would probably have the most competition with the small tree finch, should they live on one island? Give a reason for your answer. (3)

[9]

3.3. Look at the graph below and explain the trend that has occurred in the human population growth. Give THREE possible reasons for this growth pattern. [5]

HUMAN POPULATION GROWTH



Total Question 3: 30 Marks

Question 4:

Question:

A patient diagnosed with Multiple Sclerosis (MS) can be prescribed drugs to help slow the progression of the condition. Can these drugs affect the normal conduction of the nervous system?

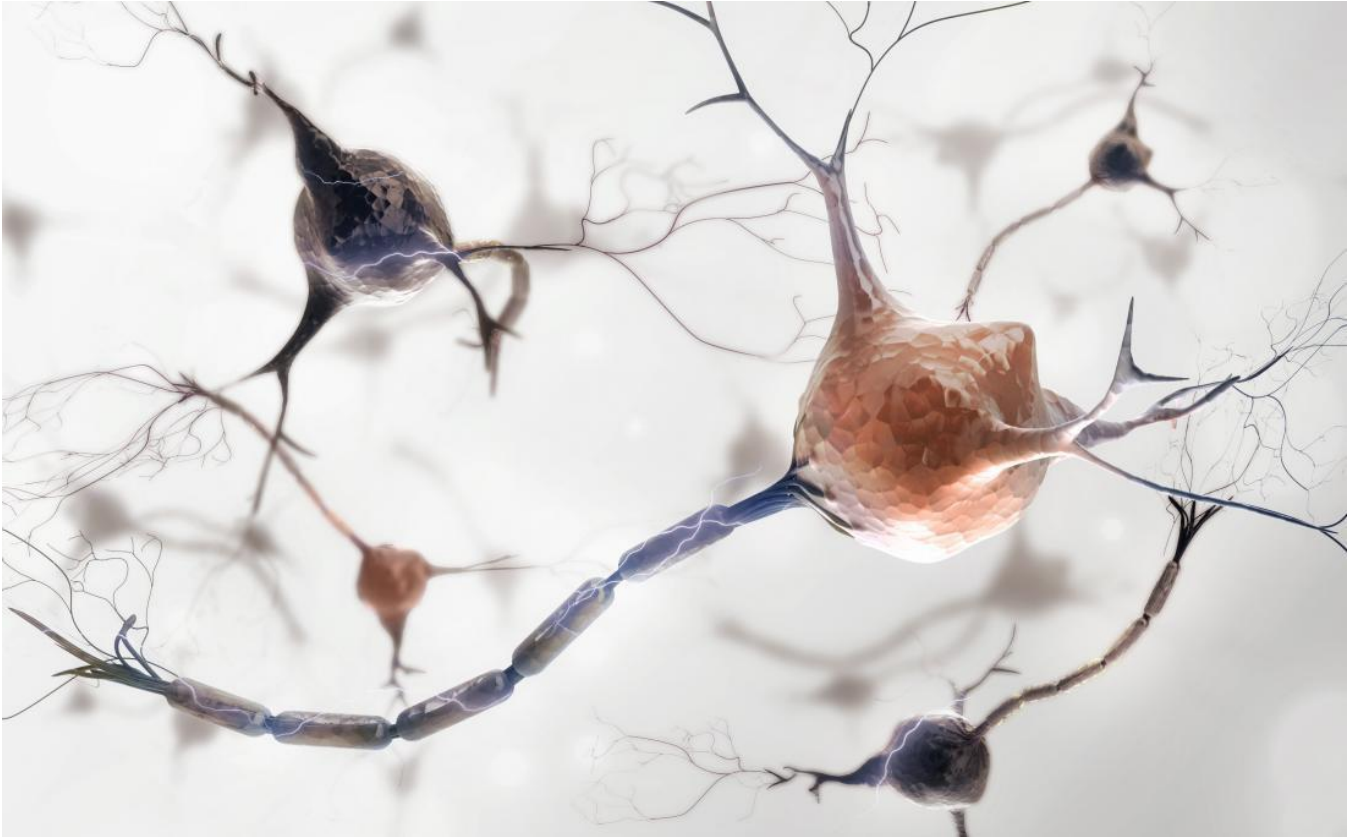
Using the source material provided as well as your own knowledge, discuss your opinion on the above statement in the form of a 2½–3 page essay.

To answer this question you are expected to:

- Read the source material carefully and present a debated argument to illustrate your point of view.
- Select relevant information from sources A to G below.
- It is important to integrate your own relevant biological knowledge.
- Take a definite stand on the question and arrange the information to best develop your argument.
- Write in a way that is scientifically appropriate and communicates your point of view clearly.
- Provide a clear plan of your essay before you start writing. Note that the plan will be marked as part of the assessment of this question.

Source A:

Multiple sclerosis is an often debilitating autoimmune disease characterized by muscle weakness, vision problems, impaired coordination, and physical numbness. Currently, treatments for this disease focus on managing the symptoms, but what if we could tackle their main cause in the brain?



Are we getting closer to learning how to prompt cells to fix the neural 'sheath' damage at the center of MS? Although the root causes of multiple sclerosis(MS) remain unknown, we do know that one factor for its symptoms is demyelination.

During this process, the immune system attacks and damages myelin, or the "coating" around axons, which are the projections that connect nerve cells in the brain and spinal cord. As a result of this damage, signals transmitted between nerve cells are disrupted, hence the problems with vision, coordination, or muscle control. Therefore, the main challenge for researchers specializing in MS has been how to promote remyelination, which is the creation of a new myelin "sheath," in an efficient and effective way.

According to a 2016 study, more than 403,600 people in the United States live with MS, while an earlier study pointed to an estimated number of 2.3 million people diagnosed with the condition around the world.

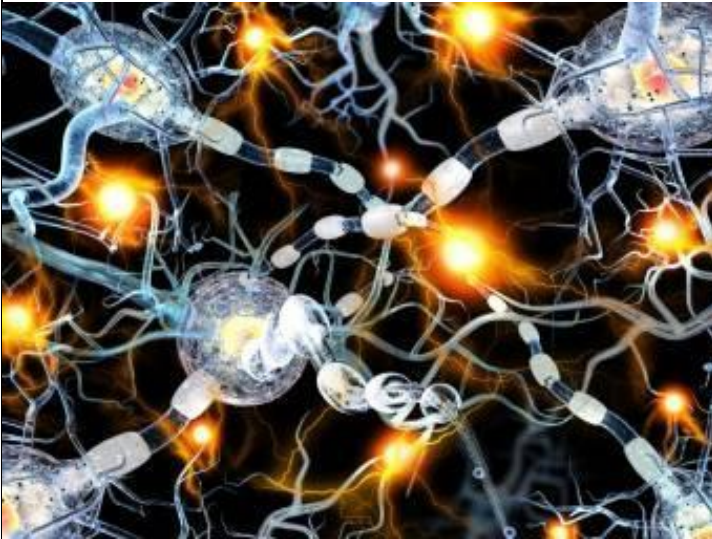
Dr. Veronique Miron — at the MS Society Edinburgh Centre for MS Research in the United Kingdom — and other scientists made a breakthrough in remyelination research when they found that a protein called activin-A plays an important role in the promotion of myelin repair. At the time, they could not

identify the mechanism whereby the protein boosts myelination. But now, Dr. Miron and team have discovered how this protein "switches on" the process of repair.

'Encouraging cells to make new myelin'

Dr. Miron and colleagues studied the myelin production mechanism in which activin-A is implicated both in vivo (using the mouse model of MS) and in vitro (on human tissue provided by the [MS Society Tissue Bank](#)).

The scientists found that the processes that led to the production of myelin were dependent on the expression of an activin-A receptor called activin receptor 2a (Acvr2a) on oligodendrocytes, a type of cell able to create myelin.



Multiple sclerosis: 'Guardian molecule' may lead to new treatment

Looking at tissue samples donated by people who had lived with progressive MS, Dr. Miron and her team saw that Acvr2a levels were significantly higher in nervous tissue that benefitted from remyelination. By contrast, Acvr2a levels were reduced in tissue with no signs of myelin repair. Activin-A, the researchers also discovered, binds to Acvr2a, signaling the oligodendrocytes to start their work of repair at sites where the axons' myelin sheath has been damaged.

"When we first discovered this protein activin-A," Dr. Miron notes, "we didn't know exactly what role it played in remyelination. We now know it binds to a specific receptor, which then causes cells to carry out myelin repair."

As re-myelination is crucial to slowing down the progression of the disease and potentially halting it altogether, Dr. Miron says that the findings of the current study could eventually lead to a new drug target.

Adapted from: <https://www.medicalnewstoday.com/articles/320859.php>

Source B: Disease-Modifying Drugs

There's no cure for multiple sclerosis (MS), but medicines and lifestyle changes can help you manage the disease. Work closely with your doctor to find the treatment that's best for you and causes the fewest side effects.

If you have a type of multiple sclerosis called relapsing-remitting MS and your condition is acting up, your doctor may first treat you with a disease-modifying drug. These medicines slow down the advance of your disease and prevent flare-ups.

The drugs work by curbing the immune system -- your body's main defense against germs -- so that it doesn't attack the protective coating called myelin that surrounds the nerves.

Some drugs come as injections administered under your skin or into a muscle. The shot might make your skin sore, red, itchy, or dimply. They include:

Beta interferons: These are some of the most common drugs used to treat MS. They ease the severity and frequency of flares. They can also cause flu-like symptoms, like aches, fatigue, fever, and chills, but these should fade within a few months. They may make you slightly more likely to get an infection. That's because they lower the number of white blood cells, which help your immune system fight illnesses.

Adapted from: <https://www.webmd.com/multiple-sclerosis/ms-treatment#1>

Source C: Scott Mills: It's wonderful to finally have a drug that could help mum's MS

Since his mum Sandra's devastating diagnosis 11 years ago, when she was 54, the 44-year-old, who has also appeared on Casualty and Strictly Come Dancing, has worked tirelessly to raise the profile of the disease which affects more than 100,000 people in the UK.

So when it was announced that a drug called ocrelizumab has been approved by the European Medicines Agency as the first licensed treatment for those with primary progressive MS (PPMS), after trials showed it slowed progression of the disease, Scott and fellow campaigners were delighted.

Ocrelizumab, which is taken as an intravenous infusion every six months, is currently being appraised by The National Institute for Health and Care Excellence (NICE) and the Scottish Medicines Consortium (SMC) to determine whether it will be made available on the NHS. A decision is expected in July this year, with the drug potentially available on the NHS in England and Wales three months later.

"It's wonderful to have something at last," says Scott, who lives in London.



Adapted from: <https://www.express.co.uk/Scott-Mills-mother-Sandra-Multiple-Sclerosis-sufferer-new-drug-Ocrelizumab>

Source D:

Mind-altering drugs may slow down or speed up the central nervous system and autonomic functions necessary for living, such as blood pressure, respiration, heart rate, and body temperature. Levels of some of the brain's chemical messengers, or neurotransmitters, are also impacted by drug abuse, including:

- Dopamine: This neurotransmitter regulates moods, enhances pleasure, and is involved with movement, reward and reinforcing behaviours, motivation, and attention.
- Serotonin: This neurotransmitter is responsible for stabilizing moods and regulating emotions.
- Gamma-aminobutyric acid (GABA): GABA acts as a natural tranquilizer, mitigating the stress response and lowering anxiety levels as well as slowing down functions of the central nervous system.
- Norepinephrine: Similar to adrenaline, norepinephrine is often called the "stress hormone," as it speeds up the central nervous system in response to the "fight-or-flight" response. It also homes focus and attention while increasing energy levels

Regions of the brain are disrupted by drug abuse, as the National Institute on Drug Abuse(NIDA) reports that the brain stem, limbic system, and cerebral cortex are all affected. The brain stem controls life-sustaining functions, including sleeping, breathing, and heart rate, while the limbic system holds the brain's reward circuitry and helps to control emotions and the ability to feel happiness. The cerebral cortex is considered the "thinking center" of the brain, managing problem-solving, planning, and decision-making abilities as well as helping people to process information provided by their senses.

The American Society of Addiction Medicine (ASAM) defines addiction as a disease affecting brain chemistry and circuitry, which then leads to compulsive drug-seeking and using behaviors. In 2014, nearly 22 million Americans battled addiction, NSDUH reports.

Adapted from: <https://americanaddictioncenters.org/central-nervous-system/>

Source E: Drugs Disrupt Neurotransmission

How do drugs cause their effects on the brain and behavior? Lesson 1 introduced students to the idea that a specific brain region, the reward system (part of the limbic system), regulates feelings of pleasure and that this region is activated by drugs of abuse. But what do drugs actually do in that brain region? Drugs interfere with neurotransmission. More specifically, drugs of abuse produce feelings of pleasure by altering neurotransmission by neurons in the reward system that release the neurotransmitter dopamine.

Thus, drugs of abuse alter the communication between neurons that is mediated by dopamine. Because the synapse is so complex, there is a variety of sites at which drugs may affect synaptic transmission. One way to affect synaptic transmission is to increase the amount of neurotransmitter released into the synaptic space. Drugs like alcohol, heroin, and nicotine indirectly excite the dopamine-containing neurons in the ventral tegmental area (VTA) so that they produce more action potentials. As the number of action potentials increases, so does the amount of dopamine released into the synapse. Amphetamines (e.g., methamphetamine, crystal, crank) actually cause the release of dopamine from the vesicles. This is independent of the rate of action potentials and, depending on dose, can cause a relatively quick and prolonged rise of extracellular dopamine levels.

Drugs may also alter synaptic transmission by directly affecting the postsynaptic receptors. Some drugs activate receptors, and others block them.

While THC (the main psychoactive chemical in marijuana) and morphine activate specific receptors, other drugs block specific receptors. Caffeine, the mild stimulant found in coffee and some soft drinks, exerts its effects by preventing a neurotransmitter/neuromodulator called adenosine from binding to its receptor. Normally, the binding of adenosine to its receptor causes sedation; it is a natural sleep-inducer. Instead of causing sedation, the blocking of the adenosine receptors with caffeine leads to an increase in activity and arousal levels.

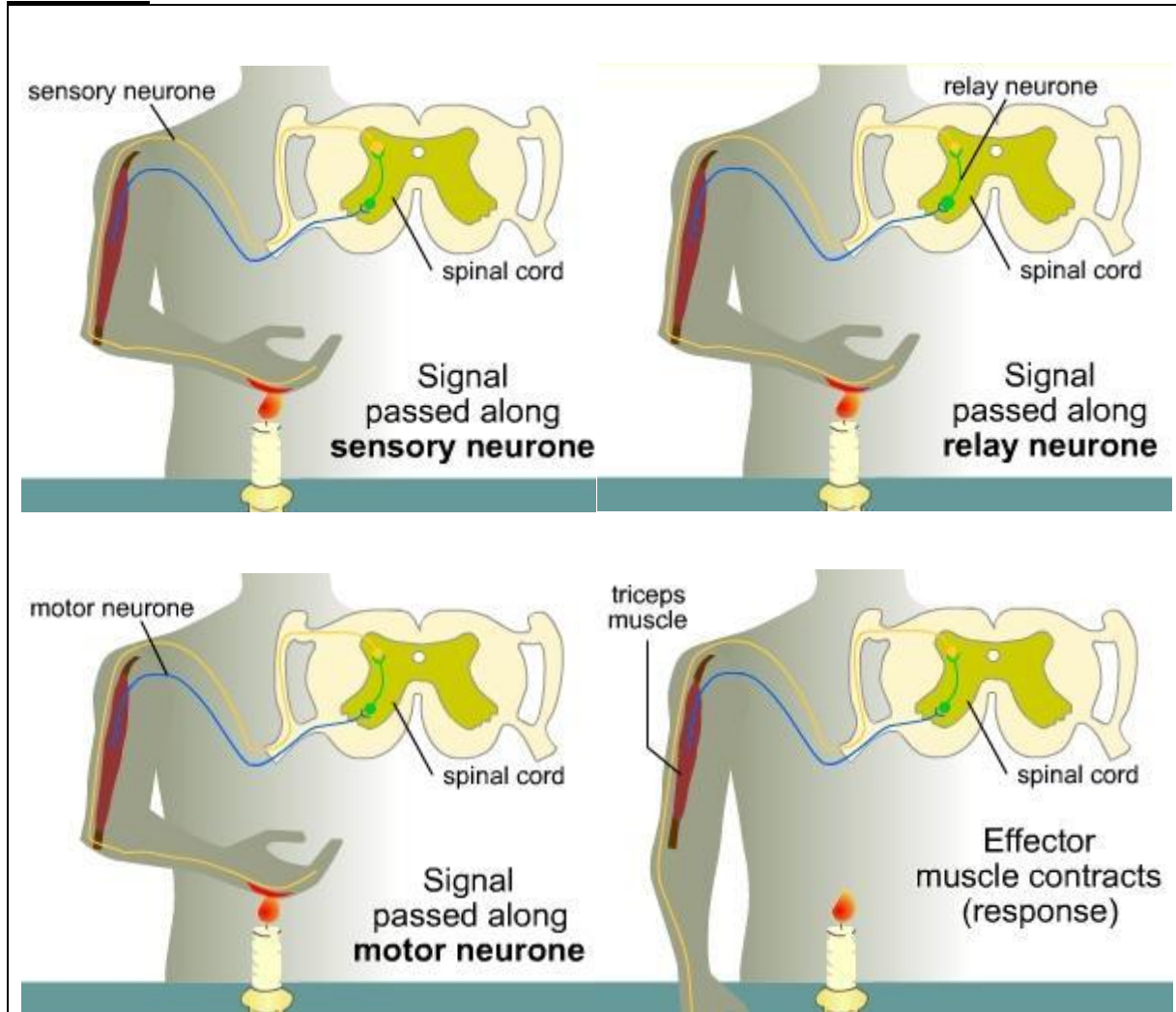
The actions of some drugs are very complex. LSD, for example, acts on serotonin receptors. Serotonin, an important neurotransmitter in many brain regions, is involved in regulating a wide variety of functions, including mood and basic survival functions such as sleeping and eating. Scientists continue to study how hallucinogens act, but apparently LSD activates some serotonin receptors (LSD acts as a receptor agonist) and blocks other serotonin receptors (LSD acts as a receptor antagonist).

A third way to affect synaptic transmission is to alter the removal of neurotransmitters from the synapse. Cocaine and amphetamines work this way (this is the second way amphetamines can alter neurotransmission).

Both drugs block the dopamine transporter (reuptake pump) that removes dopamine from the synapse. The result is a fairly rapid and persistent rise of dopamine in the synapse, leading to feelings of euphoria and well-being. Most drugs of abuse don't block enzymatic destruction of neurotransmitters, although smoking has been shown to reduce levels of an enzyme that breaks down neurotransmitters, monoamineoxidase.

Adapted from: <https://science.education.nih.gov/supplements/webversions/BrainAddiction/guide/lesson3-1.html>

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Source F:

<http://www.frankswebpace.org.uk/ScienceAndMaths/biology/reflex-actions.htm>

SOURCE G:**Who is at risk of MS?**

Research suggests that risk factors of MS include a lack of vitamin D or sunlight, smoking, obesity, a history of infection with the Epstein-Barr virus, being female, and possibly having inherited specific genes, as well as environmental factors. Researchers speculate that balancing out the populations of gut bacteria in people with MS may reduce inflammation and the over activation of the immune system.

What are the latest treatments for MS?

DMARDs that health authorities have recently approved as MS treatments include cladribine (Mavenclad) and siponimod (Mayzent) for relapsing-remitting and active secondary progressive forms of the condition.

Cladribine targets lymphocytes, white blood cells responsible for attacks on myelin. Siponimid harnesses specific white blood cells that attack they myelin and prevents them from circulating in in the body.

However, due to their interactions with the immune system, these drugs may lead to a reduction in lymphocytes, making a person vulnerable to infections. The medicines' actions may also contribute to reduced responses to vaccines in people who receive routine vaccinations. With the introduction of COVID-19 vaccines, scientists have investigated whether people with MS who take medications such as cladribine can have adequate responses to vaccines.

The latest research indicates that people taking cladribine do produce protective antibodies to other common vaccines, despite having decreased lymphocyte levels induced by the medication. This result gives scientists and others in the medical community hope that people who take these drugs for MS will have similarly adequate responses to COVID-19 vaccines.

Research on the horizon

Some scientists are currently investigating the potential for stem cell therapy for MS. In a phase 1 study conducted at the Karolinska Institute, in Stockholm, Sweden, seven people with progressive MS received infusions of stem cells derived from each participant's own bone marrow.

As early as 7 days after administration of the stem cell therapy, researchers found evidence of positive changes in the participants' immune systems. At 12 weeks, five out of six participants had no new characteristic lesions on follow-up MRI brain scans.

As their understanding of the condition evolves, many scientists are investigating the root cause of MS. An analysis of the current data has revealed a possible connection between gut health and the condition. Data revealing relationships between the gut microbiota and the brain continually emerge, and scientists are hopeful that diet modifications, probiotics, and certain drugs that balance the gut microbiome will play a role in MS treatment. Also in development are remyelination and neuroprotection therapies. The latter aim to protect the axons and myelin from further damage, while the former could restore lost function for people with MS.

Meanwhile, immunotherapy drugs would protect the nerves from destruction and rebuild neurons that have already sustained damage. Another potential treatment in phase 1 trials is a tumor necrosis factor-alpha (TNF-alpha) inhibitor called MYMD-1. TNF-alpha is a type of cytokine produced by white blood cells that regulates some aspects of the immune system. Overproduction of this cytokine is associated with several autoimmune conditions, including MS. MYMD-1 is a new type of TNF-alpha blocker that shows promise as a treatment for MS and other conditions.

Trials for therapies involving the gut microbiome, stem cells, neuroprotective treatments, remyelination, and MYMD-1 are still in the earliest stages. However, the possibilities provide hope that ongoing research will lead to effective ways to prevent MS and better methods of treatment.

Adapted from: <https://www.medicalnewstoday.com/articles>

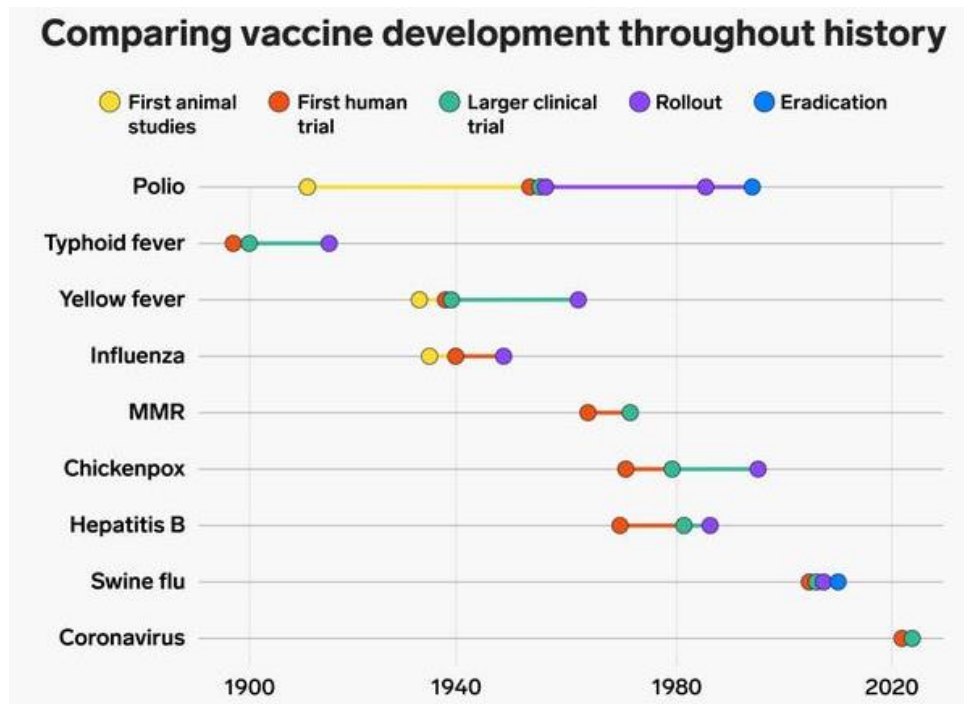
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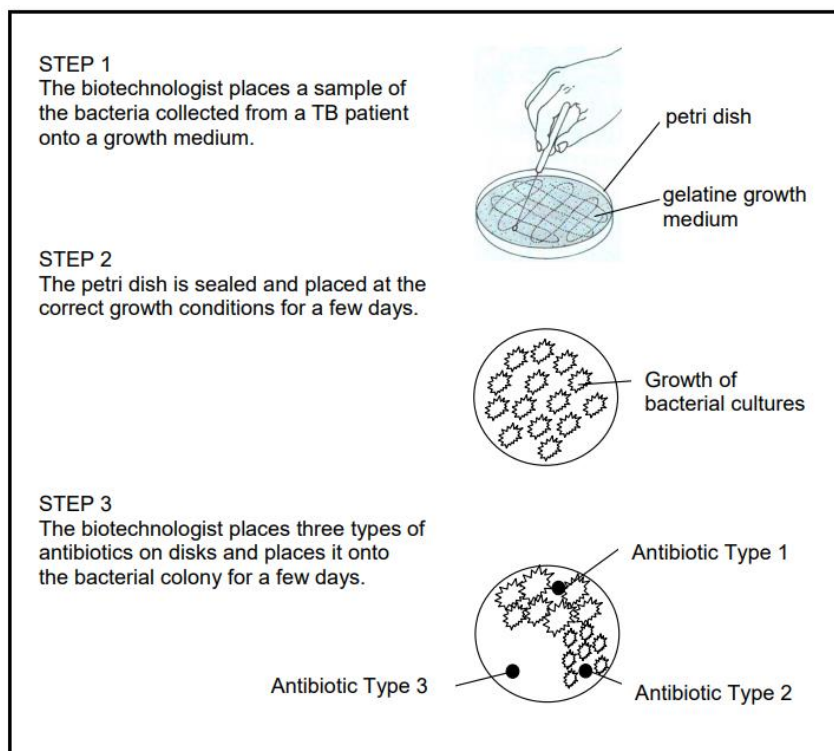
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Appendix

Question 2.1.5.



Question 2.2.





Question 4: Source F

